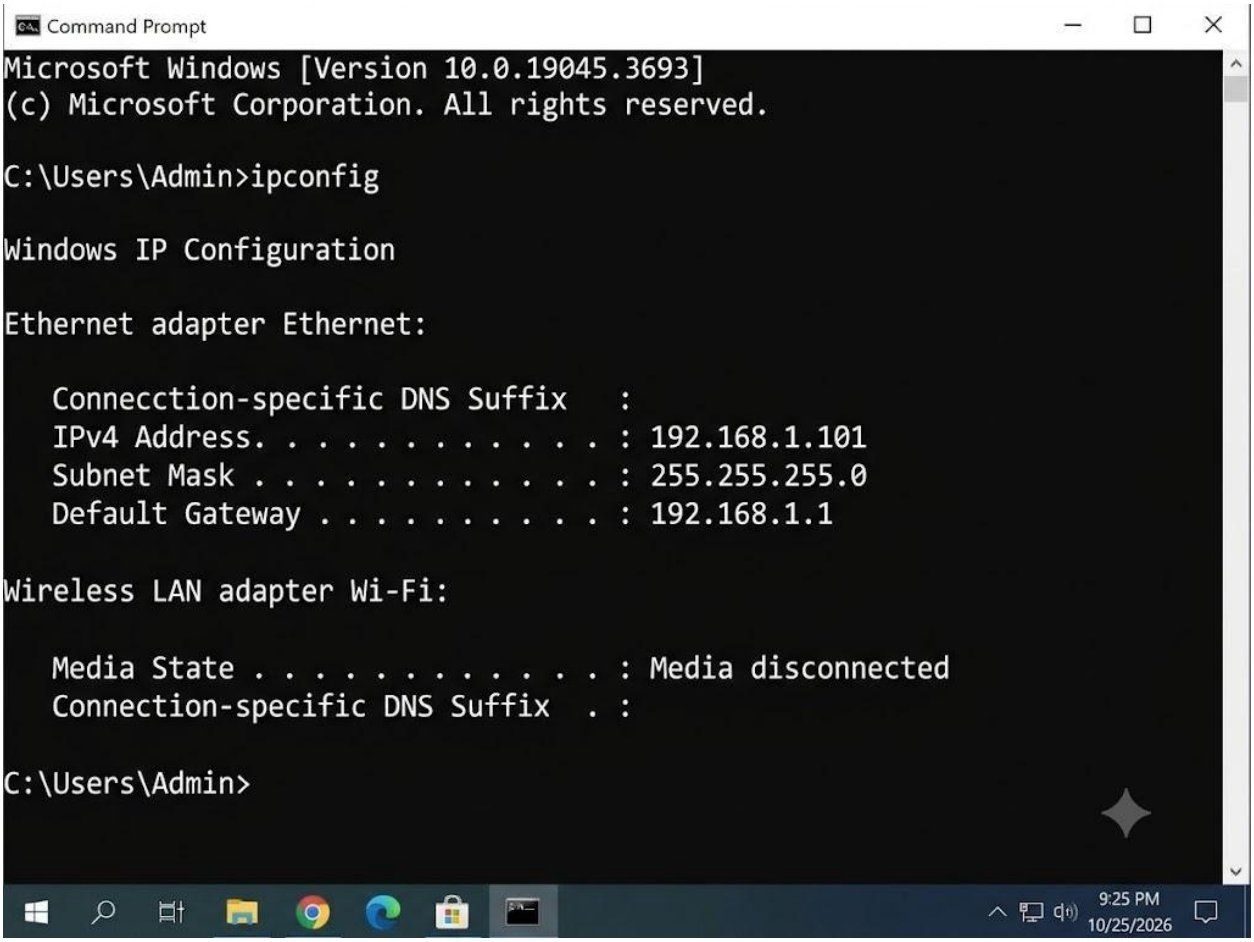
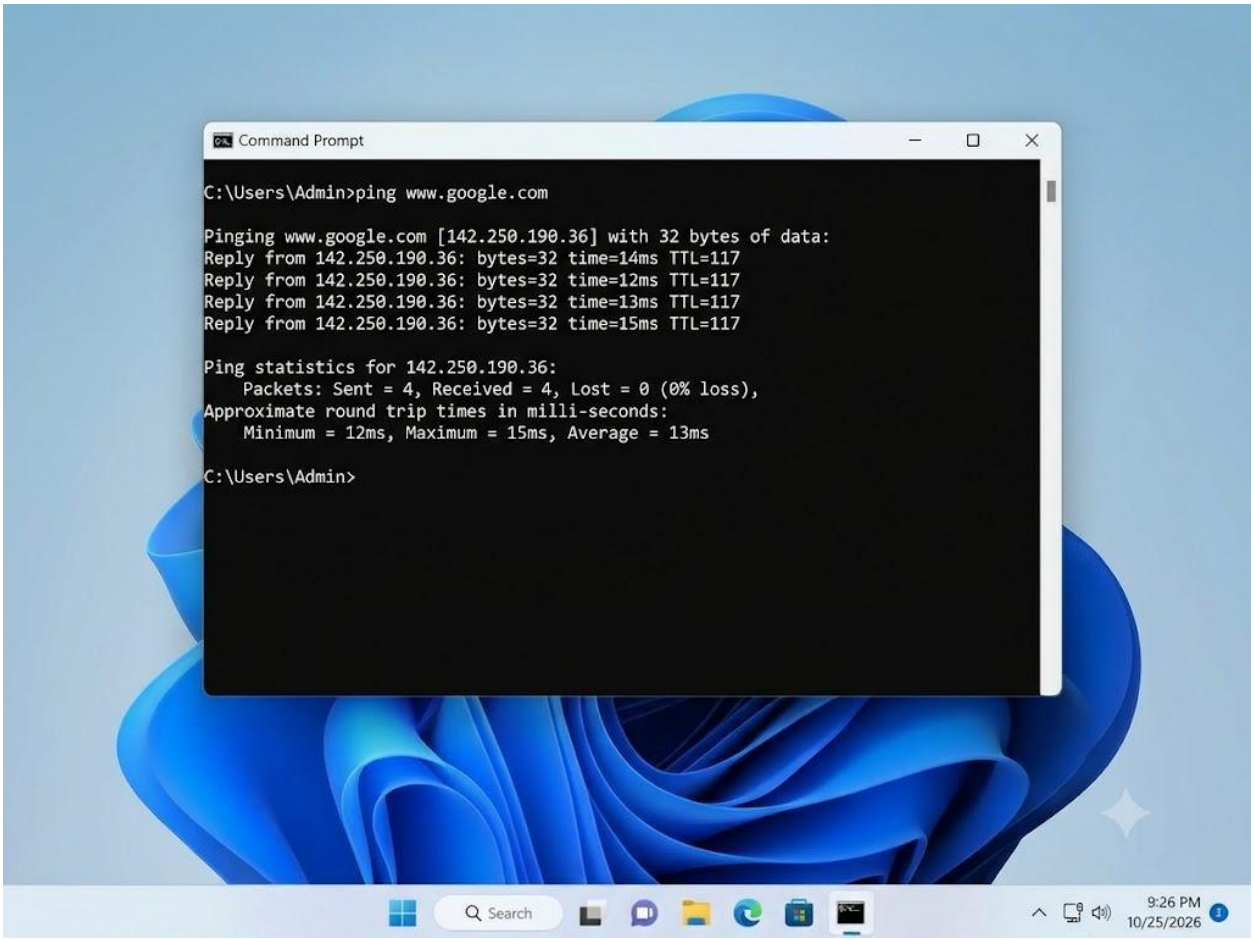
1.

Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.



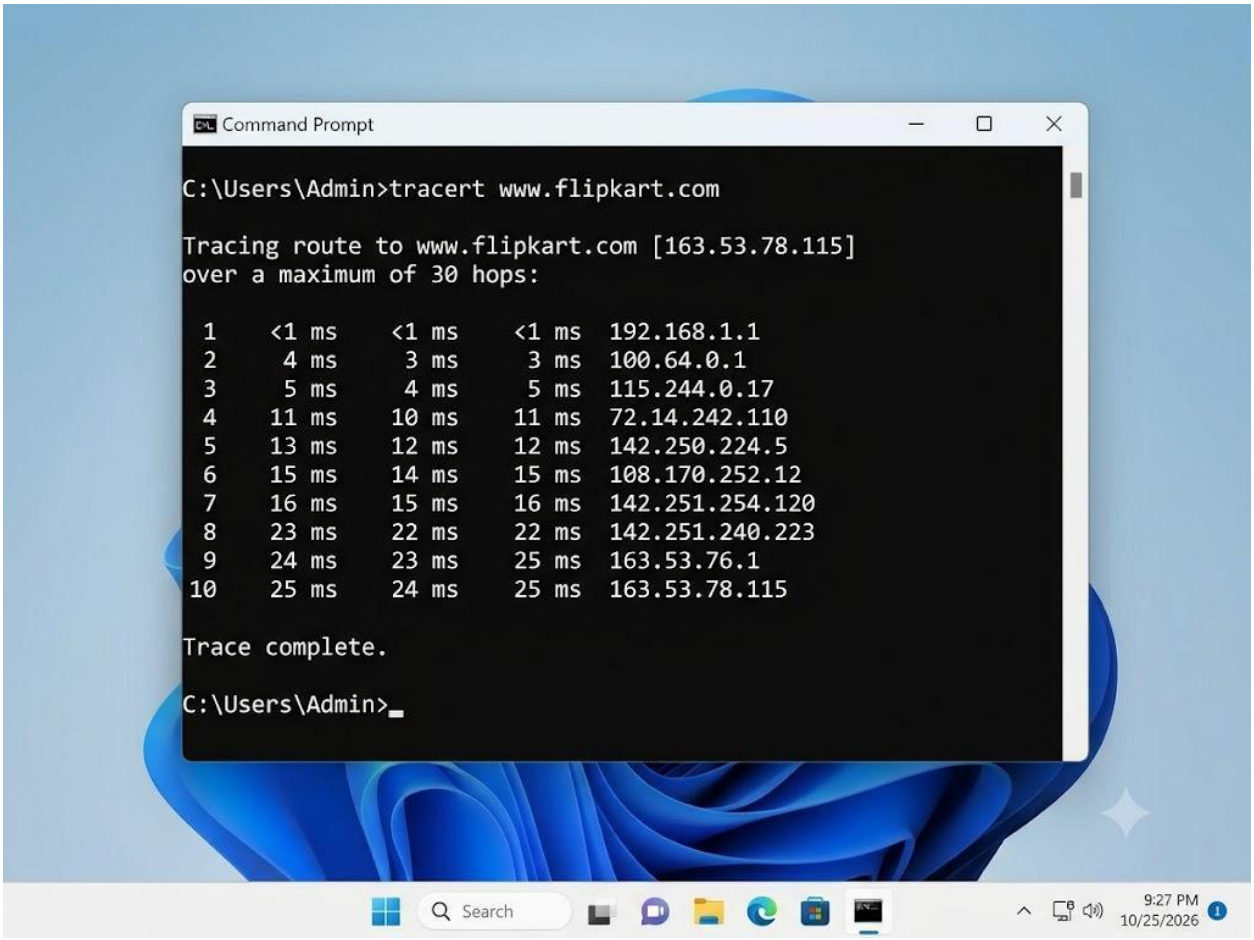
2.

Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.



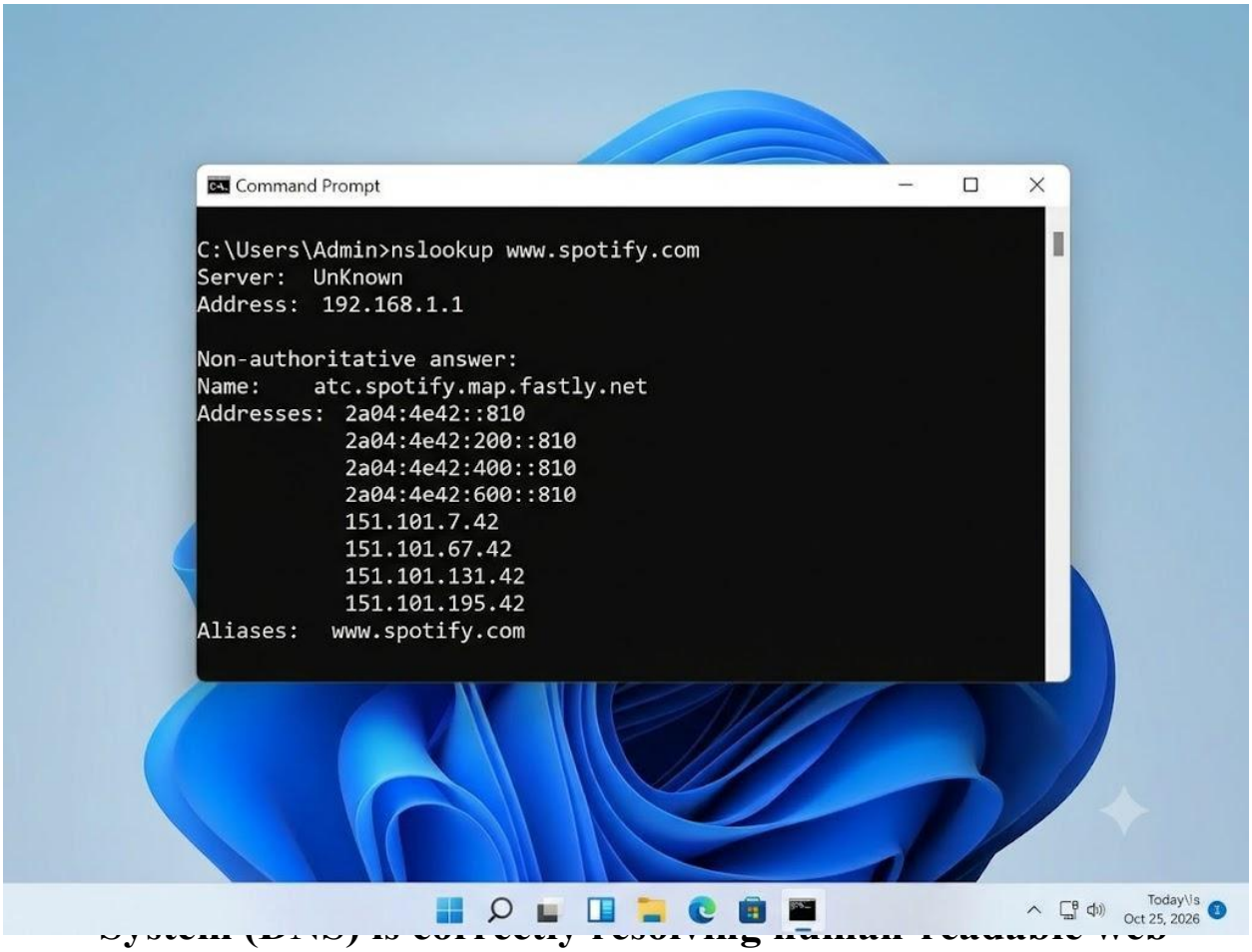
3.

Run the tracert (or traceroute on Linux/Mac) command to www.flipkart.com and list the number of hops it takes to reach the destination.



4.

Use the nslookup tool to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.



System (DNS) is correctly resolving human-readable web addresses into the IP addresses needed for your network to communicate with them.

5.

A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check. Hint: Think about checking both network connectivity and DNS resolution.

1. **ipconfig:** Checks the laptop's IP address, subnet mask, and default gateway to verify network configuration.

2. **nslookup zomato.com:** Checks whether DNS can resolve Zomato's domain name to an IP address.

3. **ping zomato.com:** Tests whether the laptop can reach Zomato's server, although some servers may block ping requests.

Result: These three commands help identify IP configuration, DNS resolution, and network connectivity problems.